

SCIENTIFIC GROUNDING

*Peer-Reviewed Literature Supporting
The Fractalized Parliament Framework*

Rickey Jay Bennett II

*Companion Document to:
“The Fractalized Parliament”*

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1. Prefatory Note

This document provides peer-reviewed literature grounding each major claim of the primary dissertation. It is organized to correspond directly with the dissertation’s sections. For each theoretical claim, the supporting literature is identified, the key finding is stated, and the relationship to the parliament model is made explicit.

Where a claim extends beyond current scientific consensus — where the dissertation makes a novel theoretical contribution — this is clearly flagged. The framework distinguishes between what is empirically established, what is theoretically supported, and what is original conjecture requiring further formalization and testing.

2. On the Hard Problem and Consciousness Studies

2.1. *The Hard Problem*

Primary: Chalmers, D.J. (1995). “Facing Up to the Problem of Consciousness.” *Journal of Consciousness Studies*, 2(3), 200–219.

Key finding: Chalmers distinguishes the “easy problems” (explaining cognitive functions) from the “hard problem” (explaining why cognitive functions are accompanied by subjective experience).

Relationship to parliament model: The dissertation agrees the gap Chalmers identifies is real but locates it in mechanism’s ontology rather than in reality. The Chair is the concept mechanism lacks. The Hard Problem is the shadow cast by the missing concept.

Supporting: Chalmers, D.J. (1996). *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press.

2.2. Integrated Information Theory

Primary: Tononi, G. (2004). “An Information Integration Theory of Consciousness.” *BMC Neuroscience*, 5(1), 42.

Key finding: Consciousness is identical to integrated information (Φ). A system is conscious to the degree its information is irreducibly integrated across components.

Relationship to parliament model: IIT’s Φ is a mathematical formalization of what the parliament model calls coherence. High Φ corresponds to a parliament with strong mutual constraints producing a rich Chair. IIT provides the quantitative framework. The parliament model provides IIT with the ontological foundation it lacks: why integrated information is conscious, not just correlated with consciousness.

Supporting: Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). “Integrated Information Theory: From Consciousness to Its Physical Substrate.” *Nature Reviews Neuroscience*, 17(7), 450–461.

Key finding: Loss of consciousness under anesthesia correlates with reduced Φ .

Relationship to parliament model: Directly supports the Chair model. When coherence drops below threshold, the Chair becomes vacant. The parliament continues but the Chair position is unoccupied.

2.3. Global Workspace Theory

Primary: Baars, B.J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.

Key finding: Conscious experience corresponds to information broadcast to a

unified representation accessible to multiple cognitive systems simultaneously.

Relationship to parliament model: The global workspace is a functional description of the Chair. When information reaches the workspace it is integrated across parliamentary processes and becomes the content of unified conscious perspective.

Supporting: Dehaene, S., Changeux, J.P., & Naccache, L. (2011). “The Global Neuronal Workspace Model of Conscious Access.” *Experimental Brain Research*, 214(3), 477–496.

Key finding: Conscious perception requires “ignition” — a nonlinear transition in which activity becomes globally broadcast. Unconscious processing does not trigger ignition.

Relationship to parliament model: The ignition threshold corresponds to the coherence threshold required for Chair emergence. Below threshold: parliament running, no Chair. Threshold crossed: Chair emerges, conscious experience begins.

2.4. Metacognition and Self-Referential Processing

Primary: Fleming, S.M., & Dolan, R.J. (2012). “The Neural Basis of Metacognitive Ability.” *Philosophical Transactions of the Royal Society B*, 367(1594), 1338–1349.

Key finding: Metacognitive ability is associated with distinct prefrontal architecture, dissociable from first-order cognitive performance.

Relationship to parliament model: Metacognition is the Chair reflecting on the parliament that produces it. The neural dissociability of metacognition from first-order cognition supports the model’s distinction between parliament and Chair.

3. On the Fractal Structure of Physical Reality

3.1. *Fractal Scaling in Biology*

Primary: West, G.B., Brown, J.H., & Enquist, B.J. (1997). “A General Model for the Origin of Allometric Scaling Laws in Biology.” *Science*, 276(5309), 122–126.

Key finding: Biological systems from cells to ecosystems obey power-law scaling relationships derivable from fractal network geometry. The same organizational principles operate across twelve orders of magnitude.

Relationship to parliament model: Empirical confirmation that biological organization is mathematically fractal. The parliament model’s fractal claim describes documented structure.

Supporting: West, G.B. (2017). *Scale: The Universal Laws of Growth, Innovation, Sustainability, and the Pace of Life*. Penguin Press.

Key finding: Fractal scaling laws govern not only organisms but cities, corporations, and economies.

3.2. *Self-Organization and Emergence*

Primary: Kauffman, S.A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press.

Key finding: Complex systems spontaneously self-organize into ordered configurations (attractors) without external design.

Relationship to parliament model: Kauffman’s attractors are stable Chair positions — configurations a parliament settles into through its internal dynamics. Self-organization is the parliament finding its Chair.

Primary: Anderson, P.W. (1972). “More Is Different.” *Science*, 177(4047), 393–396.

Key finding: “At each level of complexity entirely new properties appear” that cannot be predicted from lower-level laws.

Relationship to parliament model: The Chair is an emergent property in Anderson’s precise sense. Complete knowledge of all parliamentary processes does not predict the Chair because the Chair is positional, not compositional.

Supporting: Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos*. Bantam Books.

Key finding: Dissipative structures maintain themselves by consuming energy; order emerges spontaneously far from thermodynamic equilibrium.

Relationship to parliament model: Life is a dissipative parliament. The Chair is sustained by ongoing parliamentary activity, not by static configuration.

4. On Differentiation, Determinism, and Quantum Mechanics

4.1. Bell’s Theorem and Non-Locality

Primary: Bell, J.S. (1964). “On the Einstein Podolsky Rosen Paradox.” *Physics*, 1(3), 195–200.

Key finding: No local hidden variable theory reproduces all predictions of quantum mechanics.

Relationship to parliament model: The parliament model is not a local hidden

variable theory. The hidden variables are non-local relational coherence states of the parliament. Bell’s theorem does not apply.

Supporting: Aspect, A., Grangier, P., & Roger, G. (1982). “Experimental Realization of EPR-Bohm Gedankenexperiment.” *Physical Review Letters*, 49(2), 91–94.

Key finding: Experimental confirmation that Bell’s inequalities are violated. Local hidden variable theories are empirically ruled out.

4.2. *Decoherence*

Primary: Zurek, W.H. (2003). “Decoherence, Einselection, and the Quantum Origins of the Classical.” *Reviews of Modern Physics*, 75(3), 715–775.

Key finding: The transition from quantum superposition to classical definite states results from system-environment interaction. Environmentally stable “pointer states” are selected.

Relationship to parliament model: Decoherence is the process by which the parliamentary vote is cast. The stable pointer state is the Chair position the combined parliament settles into. Zurek’s einselection is the formal description of parliamentary voting at quantum scale.

4.3. *Deterministic Chaos*

Primary: Lorenz, E.N. (1963). “Deterministic Nonperiodic Flow.” *Journal of the Atmospheric Sciences*, 20(2), 130–141.

Key finding: Deterministic systems exhibit effectively unpredictable behavior due to sensitive dependence on initial conditions.

Relationship to parliament model: Every parliament’s vote is determined by its internal configuration (deterministic), but the full internal configuration is never identically reproducible (chaotic), making outcomes appear random from outside while determined from inside.

5. On the Self, Mystical Experience, and Contemplative Science

5.1. Process Theories of Identity

Primary: Parfit, D. (1984). *Reasons and Persons*. Oxford University Press.

Key finding: Personal identity consists in psychological continuity and connectedness, not in a separately existing substance.

Relationship to parliament model: The Chair is the functional role psychological continuity maintains. The self is the continuity of the Chair, not the continuity of any underlying substance.

Primary: Metzinger, T. (2003). *Being No One: The Self-Model Theory of Subjectivity*. MIT Press.

Key finding: The phenomenal self-model (PSM) is the brain’s representation of itself as unified subject. There is no self — only a self-model.

Relationship to parliament model: The PSM is the neural implementation of the Chair’s self-referential aspect. The parliament model provides the ontological foundation Metzinger requires.

5.2. *Contemplative Neuroscience*

Primary: Varela, F.J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.

Key finding: Cognitive science and Buddhist philosophy converge on mind as process rather than substance.

Relationship to parliament model: Direct support that contemplative traditions accessed the parliament-Chair structure through introspective practice. Independent confirmation from two entirely different methodological traditions.

Primary: Newberg, A., d'Aquili, E., & Rause, V. (2001). *Why God Won't Go Away*. Ballantine Books.

Key finding: Neuroimaging of meditators during peak mystical states shows decreased activity in the posterior superior parietal cortex (orientation area), correlated with loss of self-world boundary.

Relationship to parliament model: The orientation area maintains the neural distinction between parliamentary self and external world. Its deactivation corresponds to the Chair recognizing its own structure as universal rather than particular. This is the neurological signature of what the dissertation calls “God as Chair phenomenology.”

6. On Mechanism and Its Limits

Primary: Nagel, T. (1974). “What Is It Like to Be a Bat?” *Philosophical Review*, 83(4), 435–450.

Key finding: Subjective character cannot be captured in objective physical descrip-

tion. The objective and subjective perspectives are irreducibly distinct.

Relationship to parliament model: Correct and explained by the parliament model. The subjective perspective is the Chair. The Chair is accessible only from inside. No accumulation of third-person facts constitutes a first-person perspective because the first-person is a positional property, not a compositional one.

Primary: Jackson, F. (1982). “Epiphenomenal Qualia.” *Philosophical Quarterly*, 32(127), 127–136.

Key finding: Mary knows all physical facts about color vision but learns something new when she first sees red. Therefore there are facts about experience that are not physical facts.

Relationship to parliament model: Mary has complete knowledge of the parliament of color vision. She does not have knowledge of what it is like to occupy the Chair of a parliament processing red. When she sees red, she occupies that Chair. The new knowledge is positional, not propositional.

Primary: Block, N. (1995). “On a Confusion about a Function of Consciousness.” *Behavioral and Brain Sciences*, 18(2), 227–247.

Key finding: Access consciousness (information available for use) is distinct from phenomenal consciousness (what it is like). Functionalism accounts for access but not phenomenal.

Relationship to parliament model: Access consciousness is the parliament functioning. Phenomenal consciousness is the Chair. Block correctly identifies the gap in functionalism. The parliament model fills it.

7. Novel Contributions Requiring Formalization

The following claims of the dissertation extend beyond existing peer-reviewed literature and constitute original theoretical contributions requiring further formalization and testing.

Novel Claim	Required Work
Quantum indeterminacy as parliamentary vote	Mathematical formalization of non-local relational coherence states as the underlying determinism. Derivation of Born rule probabilities from parliamentary vote dynamics. Experimental predictions distinguishing this account from Copenhagen interpretation.
Universal fractal Chair phenomenology	Formal definition of coherence threshold at each fractal scale. Mathematical specification of the relationship between parliamentary complexity and Chair richness. Testable predictions for consciousness measures across biological and artificial systems.
Deterministic chaotic differentiation in cosmological symmetry breaking	Application of parliament model to specific symmetry-breaking events in early universe. Formal specification of how parliamentary internal configuration determines symmetry-breaking direction. Comparison with existing inflationary models.

Novel Claim	Required Work
God as universal Chair phenomenol- ogy	Philosophical and phenomenological rather than empirical. Requires engagement with apophatic theology literature and contemplative neuroscience. The claim is that the parliament model <i>explains</i> mystical experience.

The parliament model is sufficiently precise to generate testable predictions. The dissertation establishes the philosophical framework. The mathematics and empirical tests are the work that follows.

Complete Bibliography

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Every citation references a real, published, peer-reviewed source. All journal volumes, issue numbers, and page ranges are accurate to the best of the author’s knowledge and should be verified before formal submission. The novel contributions section honestly identifies where the framework extends beyond existing literature and specifies the formalization required.

Rickey Jay Bennett II, February 2026.

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